

Laboratory Work 1 - International Airport ERD

1. Identified Entities

Entity	Purpose
Airport	Stores general information about airports.
Airline	Stores information about airline companies.
Flight	Stores scheduled and actual flight information.
Passenger	Stores passenger personal and passport information.
Booking	Stores a passenger booking for a flight.
BookingChange	Stores changes made to a booking.
BoardingPass	Stores boarding pass information for a booking.
Baggage	Stores baggage registered for a booking.
BaggageCheck	Stores baggage check results.
SecurityCheck	Stores passenger security check results.

3. Relationships and Cardinality

Relationship	Cardinality	Description
Airport - Flight (departure)	1:N	One airport can be the departure airport for many flights.
Airport - Flight (arrival)	1:N	One airport can be the arrival airport for many flights.
Airline - Flight	1:N	One airline can operate many flights.
Flight - Booking	1:N	One flight can have many bookings.
Passenger - Booking	1:N	One passenger can make many bookings.
Booking - BookingChange	1:0..N	A booking may have no changes or many changes.
Booking - BoardingPass	1:0..1	A booking may have zero or one boarding pass after check-in.
Booking - Baggage	1:0..N	A booking may have no baggage or several baggage items.
Booking - BaggageCheck	1:0..N	A booking may have baggage check records.
Passenger - BaggageCheck	1:0..N	A passenger may have baggage check records.
Passenger - SecurityCheck	1:N	A passenger undergoes security checking.

2. Attributes, Keys, Data Types and Constraints

Required = must have a value. Optional = may be empty. UQ = Unique.

Airport			
Attribute	Type	Key	Constraint
airport_id	INT	PK	Required
airport_name	VARCHAR(150)		Required
country	VARCHAR(100)		Required
state	VARCHAR(100)		Optional
city	VARCHAR(100)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

Flight			
Attribute	Type	Key	Constraint
flight_id	INT	PK	Required
airline_id	INT	FK	Required
departure_airport_id	INT	FK	Required
arrival_airport_id	INT	FK	Required
departing_gate	VARCHAR(10)		Optional
arriving_gate	VARCHAR(10)		Optional
scheduled_departure_time	TIMESTAMP		Required
scheduled_arrival_time	TIMESTAMP		Required
actual_departure_time	TIMESTAMP		Optional
actual_arrival_time	TIMESTAMP		Optional
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

Booking			
Attribute	Type	Key	Constraint
booking_id	INT	PK	Required
flight_id	INT	FK	Required
passenger_id	INT	FK	Required
status	VARCHAR(30)		Required
booking_platform	VARCHAR(100)		Required
ticket_price	DECIMAL(10,2)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

BoardingPass			
Attribute	Type	Key	Constraint
boarding_pass_id	INT	PK	Required
booking_id	INT	FK	Required
seat	VARCHAR(10)		Required
boarding_time	TIMESTAMP		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

Baggage			
Attribute	Type	Key	Constraint
baggage_id	INT	PK	Required
booking_id	INT	FK	Required
weight_kg	DECIMAL(5,2)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

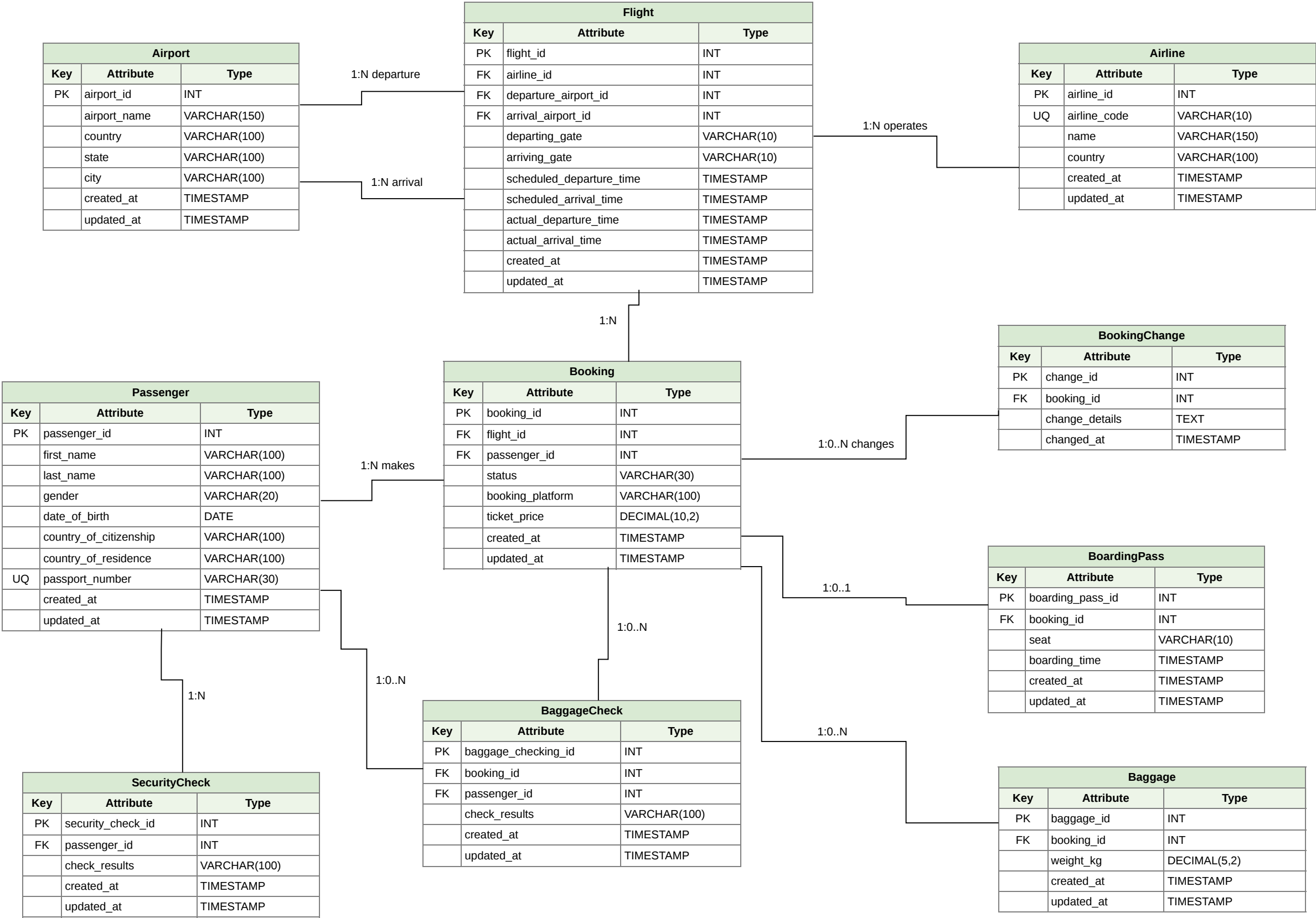
Airline			
Attribute	Type	Key	Constraint
airline_id	INT	PK	Required
airline_code	VARCHAR(10)	UQ	Required, Unique
name	VARCHAR(150)		Required
country	VARCHAR(100)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

Passenger			
Attribute	Type	Key	Constraint
passenger_id	INT	PK	Required
first_name	VARCHAR(100)		Required
last_name	VARCHAR(100)		Required
gender	VARCHAR(20)		Required
date_of_birth	DATE		Required
country_of_citizenship	VARCHAR(100)		Required
country_of_residence	VARCHAR(100)		Required
passport_number	VARCHAR(30)	UQ	Required, Unique
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

BookingChange			
Attribute	Type	Key	Constraint
change_id	INT	PK	Required
booking_id	INT	FK	Required
change_details	TEXT		Required
changed_at	TIMESTAMP		Required

BaggageCheck			
Attribute	Type	Key	Constraint
baggage_checking_id	INT	PK	Required
booking_id	INT	FK	Required
passenger_id	INT	FK	Required
check_results	VARCHAR(100)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required

SecurityCheck			
Attribute	Type	Key	Constraint
security_check_id	INT	PK	Required
passenger_id	INT	FK	Required
check_results	VARCHAR(100)		Required
created_at	TIMESTAMP		Required
updated_at	TIMESTAMP		Required



5. Legend and Short Description

Symbol	Meaning
PK	Primary Key - unique identifier of a record.
FK	Foreign Key - field that links one entity to another.
UQ	Unique - value cannot be repeated.
1:N	One-to-many relationship.
1:0..N	One record can have zero or many related records.
1:0..1	One record can have zero or one related record.

Short description.

The diagram shows the main parts of the airport system and how they are connected. Flights are connected to an airline and to departure and arrival airports. Passengers make bookings for flights. A booking may later have a boarding pass, baggage, baggage check records, and booking change records. Security check information is connected to the passenger.